



NORTH SYDNEY BOYS HIGH SCHOOL

2016 YEAR 12 ASSESSMENT TASK 1

Mathematics Extension 1

General Instructions

- **Working time – 60 minutes**
- **Reading time – 5 minutes**
- Answer all questions on the lined paper in the booklet provided.
- Write using blue or black pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Each new question is to be started on a **new page** with the exception of the multiple choice which can be answered on the same page.
- Attempt all questions.
- Use the reference sheet provided.

Class Teacher: Please tick

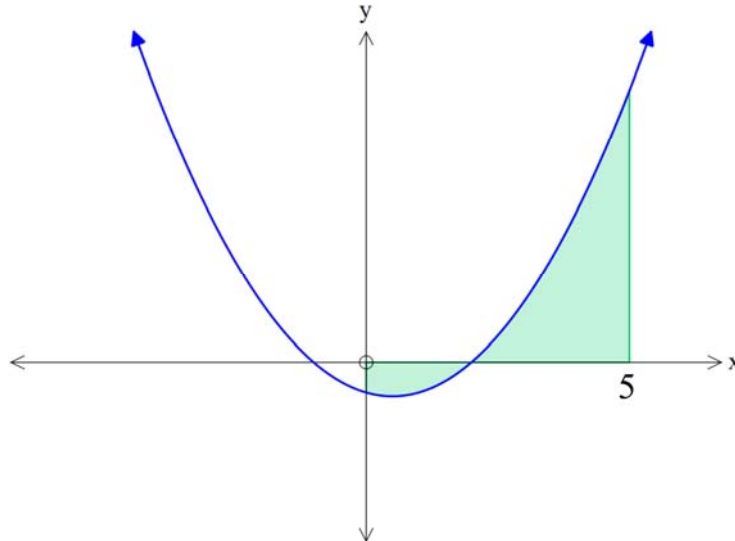
- ☐ Mr Berry
- ☐ Mr Ireland
- ☐ Mr Jomaa
- ☐ Ms Lee
- ☐ Mr Lin
- ☐ Ms Ziazaris

Student No./Name: _____

(To be used by the exam markers only.)

Section	A	B	C	D	E	F	Total	%
Mark	$\frac{\quad}{3}$	$\frac{\quad}{7}$	$\frac{\quad}{9}$	$\frac{\quad}{11}$	$\frac{\quad}{8}$	$\frac{\quad}{11}$	$\frac{\quad}{49}$	$\frac{\quad}{100}$

1. Which expression will give the area of the shaded region bounded by the curve $y = x^2 - x - 2$, the x -axis and the lines $x = 0$ and $x = 5$? 1



- (A) $Area = \left| \int_0^1 (x^2 - x - 2) dx \right| + \int_1^5 (x^2 - x - 2) dx$
- (B) $Area = \int_0^1 (x^2 - x - 2) dx + \left| \int_1^5 (x^2 - x - 2) dx \right|$
- (C) $Area = \left| \int_0^2 (x^2 - x - 2) dx \right| + \int_2^5 (x^2 - x - 2) dx$
- (D) $Area = \int_0^2 (x^2 - x - 2) dx + \left| \int_2^5 (x^2 - x - 2) dx \right|$
2. Which of the following is an expression for $\int (2x + 1)^{10} dx$ 1
- (A) $\frac{1}{9}(2x + 1)^9 + c$
- (B) $\frac{1}{11}(2x + 1)^{11} + c$
- (C) $\frac{2}{11}(2x + 1)^{11} + c$
- (D) $\frac{1}{22}(2x + 1)^{11} + c$
3. The equation of the normal to the parabola $x^2 = 4ay$ the variable point $P(2ap, ap^2)$ is given by $x + py = 2ap + ap^3$. How many different values of p are there such that the normal passes through the focus of the parabola? 1
- (A) 0
- (B) 1
- (C) 2
- (D) 3

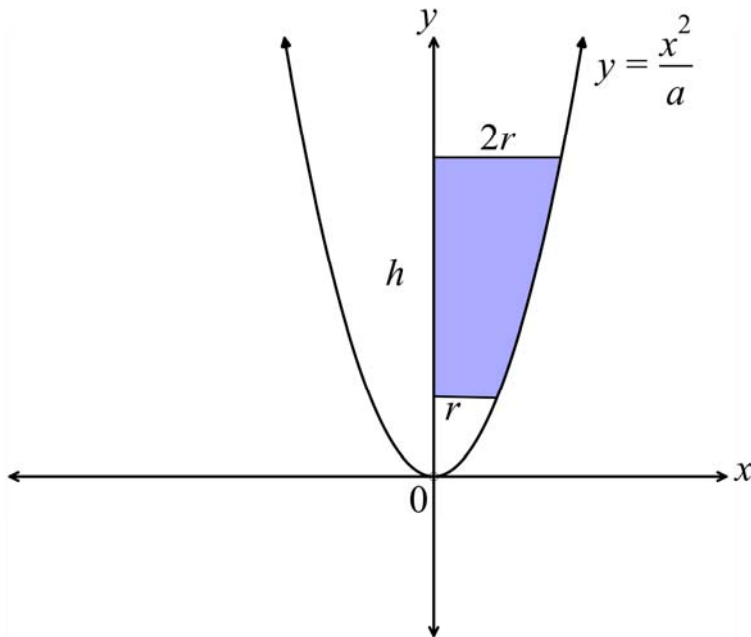
Section B (Start a new page)		Marks
1.	Find $\int \left(x + \frac{1}{x}\right)^2 dx$	2
2.	Find $\int \frac{5x^3 + 1}{x^2} dx$	2
3.	Find $\int_1^4 \sqrt{2x} dx$	3

Section C (Start a new page)		
1.	In an arithmetic sequence with $T_9 = 3T_6$ and $T_4 + T_7 = -4$. Find the value of the first term.	3
2.	Find the sum of 10 terms of the series $\log_m 3 + \log_m 6 + \log_m 12 + \dots$ given that $\log_m 3 = 0.48$ and $\log_m 2 = 0.30$	2
3.	Evaluate $\sum_{r=1}^{11} [(-1)^r 3^r + (3r + 1)]$	4

Section D (Start a new page)		
1.	The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.	
(i)	Show the equation of the normal to the parabola at P is $x + py = 2ap + ap^3$	3
(ii)	The equation of the chord PQ is $y = \frac{p+q}{2}x - apq$ (Do NOT show this.) If the chord PQ passes through the focus of the parabola, show that $pq = -1$	2
(iii)	Write down the equation of the normal to the parabola at Q . The normals intersect at N . Show that $N(-apq(p+q), a(p^2 + q^2 + pq + 2))$.	2
(iv)	If PQ is a focal chord and N is the intersection of the normals, find the equation of the locus of N and give a precise geometrical description.	4

Section E (Start a new page)		
1.	When the polynomial $P(x)$ is divided by $x^2 - 5x + 6$, the remainder is $5x + 2$. What is the remainder when $P(x)$ is divided by $x - 2$?	2
2.	The cubic equation $2x^3 - x^2 + x + 1 = 0$ has roots α, β and γ . Evaluate (i) $\alpha\beta + \beta\gamma + \alpha\gamma$ (ii) $\alpha\beta\gamma$ (iii) $\alpha^2\beta^2\gamma + \beta^2\gamma^2\alpha + \alpha^2\gamma^2\beta$	3
3.	Solve the polynomial equation $x^3 - 12x^2 + 12x + 80 = 0$ if the roots of this equation are in arithmetic progression.	3

1. Use the trapezoidal rule with five function values, to find an approximate value of the area under the curve $y = \log_{10} x$, bounded by the x -axis and $x = 1$ and $x = 3$, correct to 4 significant figures. 2
2. Consider the curves $x^2 = 4y$ and $y^2 = 4x$.
 - (i) Show that the curves intersect at $(0, 0)$ and $(4, 4)$ 1
 - (ii) Find the area enclosed between the two curves. 3
3. An open bowl with a flat circular base is made by rotating the shaded area of the curve $y = \frac{x^2}{a}$ around the y -axis. 5



The radius of the top rim of the bowl is twice that of the base, i.e. $2r$ and r respectively.

The volume of the bowl is $\frac{5}{6}\pi a^3 \text{ unit}^3$.

Find the height (h) of the bowl.

End of examination

2016 HSC Task 1 Suggested Solutions

Section A

1. C

2. D

3. B

Section B

1. $\int (x + \frac{1}{x})^2 dx$

$$= \int (x^2 + 2 + x^{-2}) dx$$

$$= \frac{x^3}{3} + 2x + (-x^{-1}) + C$$

$$= \frac{x^3}{3} + 2x - \frac{1}{x} + C$$

2. $\int \frac{5x^3 + 1}{x^2} dx$

$$= \int (5x + x^{-2}) dx$$

$$= \frac{5x^2}{2} + (-x^{-1}) + C$$

$$= \frac{5x^2}{2} - \frac{1}{x} + C$$

[2] 3. $\int_1^4 \sqrt{2x} dx$

[3]

$$= \int_1^4 (2x)^{\frac{1}{2}} dx$$

$$= \left[\frac{2\sqrt{2}x^{\frac{3}{2}}}{\frac{3}{2}} \right]_1^4$$

$$= \frac{2\sqrt{2}}{3} (8 - 1)$$

[2] $= \frac{14\sqrt{2}}{3}$

Section C

$$\text{A.P. } T_n = a + (n-1)d$$

$$1. \quad T_9 = 3T_6$$

$$T_4 + T_7 = -4$$

[3]

$$\begin{aligned} a + 8d &= 3[a + 5d] \\ &= 3a + 15d \end{aligned}$$

$$(a + 3d) + (a + 6d) = -4$$

$$2a + 7d = 0 \quad (1)$$

$$2a + 9d = -4 \quad (2)$$

$$\begin{aligned} (2) - (1) \quad 2d &= -4 \\ d &= -2 \end{aligned}$$

$$\begin{aligned} \text{Sub into (1)} \quad 2a &= 14 \\ a &= 7 \end{aligned}$$

$$2. \quad \log_m 3 + \log_m 6 + \log_m 12 + \dots$$

[2]

$$\begin{aligned} \text{A.P. } a &= \log_m 3 \quad d = \log 6 - \log_m 3 \\ &= \log_m 2 \end{aligned}$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{10} = \frac{10}{2} [2 \log_m 3 + 9 \log_m 2]$$

$$= 5 [2 \times 0.48 + 9 \times 0.30]$$

$$= 18.3$$

Section C

$$3. \sum_{r=1}^{11} [(-1)^r 3^r + (3r+1)]$$

[4]

$$r=1 \quad -3 + 4$$

$$r=2 \quad 3^2 + 7$$

$$r=3 \quad -3^3 + 10$$

⋮

G.P

A.P

$$a = -3, r = 3$$

$$a = 4, d = 3$$

$$S_n = \frac{a(r^n - 1)}{r - 1} + \frac{n}{2} [2a + (n-1)d]$$

$$S_{11} = \frac{-3((-3)^{11} - 1)}{-3 - 1} + \frac{11}{2} [2(4) + 10 \times 3]$$

$$= -132861 + 209$$

$$= -132652$$

C-3 (alt:)

1	-3 + 4	=	1
2	9 + 7	=	+ 16
3	-27 + 10	=	-17
4	+81 + 13	=	+94
5	-243 + 16	=	-227
6	+729 + 19	=	+748
7	-2187 + 22	=	-2165
8	6561 + 25	=	+6586
9	-19683 + 28	=	-19655
10	+59049 + 31	=	+59080
11	-177147 + 34	=	-177113

Section D

$$(i) \quad x^2 = 4ay$$
$$y = \frac{x^2}{4a}$$

$$y' = \frac{2x}{4a}$$
$$= \frac{x}{2a}$$

$$\text{At } P(2ap, ap^2) \quad m = \frac{2ap}{2a}$$
$$= p$$

$$m_{\perp} = -\frac{1}{p}$$

Eqn of normal at P:

$$y - y_1 = m(x - x_1)$$

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$x + py = 2ap + ap^3$$

(ii) Eqn of PQ: $y = \frac{p+q}{2}x - apq$

2

Sub $S(0, a)$ $a = \frac{p+q}{2} \times 0 - apq$

$$a = -apq$$

$$\therefore pq = -1$$

(iii) Eqn of normal at P:

2

$$x + py = 2ap + ap^3 \quad \text{--- (1)}$$

Eqn of normal at Q:

$$x + qy = 2aq + aq^3 \quad \text{--- (2)}$$

① - ②

$$py - qy = 2ap + ap^3 - 2aq - aq^3$$

$$y(p - q) = 2a(p - q) + a(p^3 - q^3)$$

$$= 2a(p - q) + a(p - q)(p^2 + pq + q^2)$$

$$y = 2a + a(p^2 + pq + q^2)$$

$$= a(2 + p^2 + pq + q^2)$$

Sub into ①

$$x = 2ap + ap^3 - py$$

$$= 2ap + ap^3 - p[a(2 + p^2 + pq + q^2)]$$

$$= 2ap + ap^3 - 2ap - ap^3 - ap^2q - apq^2$$

$$= -apq(p + q)$$

$$\therefore N(-apq(p + q), a(2 + p^2 + pq + q^2))$$

$$(iv) \quad x = -apq(p+q)$$

4

$$y = a(2 + p^2 + pq + q^2)$$

Since $pq = -1$,

$$x = a(p+q)$$

$$(p+q) = \frac{x}{a} \quad (1)$$

$$y = a(2 + p^2 - 1 + q^2)$$

$$= a(p^2 + q^2 + 1)$$

$$= a((p+q)^2 - 2pq + 1)$$

$$= a((p+q)^2 + 2 + 1)$$

$$= a((p+q)^2 + 3)$$

Sub in (1)

$$= a\left(\left(\frac{x}{a}\right)^2 + 3\right)$$

$$= \frac{x^2}{a} + 3a$$

$$x^2 = a(y - 3a)$$

Locus of N is a vertical parabola with vertex $(0, 3a)$, focal length $\frac{1}{4}a$ and is concave up.

Section E

1. $P(x) = (x^2 - 5x + 6)Q(x) + 5(x+2)$
 $P(2) = (2^2 - 5(2) + 6)Q(2) + (5(2) + 2)$
 $= 0 + 12 = 12$

2

2. $a = 2$ $b = -1$ $c = 1$ $d = 1$

3

i) $\Sigma \alpha \beta = \frac{c}{a}$

$$= \frac{1}{2}$$

ii) $\Sigma \alpha \beta \gamma = -\frac{d}{a}$

$$= -\frac{1}{2}$$

iii) $\alpha^2 \beta^2 \gamma + \beta^2 \gamma^2 \alpha + \alpha^2 \gamma^2 \beta$

$$= \alpha \beta \gamma (\alpha \beta + \beta \gamma + \alpha \gamma)$$

$$= \Sigma \alpha \beta \gamma \cdot \Sigma \alpha \beta$$

$$= \frac{1}{2} \cdot \left(-\frac{1}{2}\right)$$

$$= -\frac{1}{4}$$

Section F

1. $y = \log_{10} x$

2

x	1	1.5	2	2.5	3
y	0	$\log 1.5$	$\log 2$	$\log 2.5$	$\log 3$

$$\text{Area} = \int_1^3 \log_{10} x \, dx$$

$$= \frac{1}{2} [0 + \log 3 + 2(\log 1.5 + \log 2 + \log 2.5)]$$

$$= 0.5568 \quad (4 \text{ sig. figs.})$$

2. (i) $x^2 = 4y$

1

$$y = \frac{x^2}{4} \quad \text{--- (1)}$$

$$y^2 = 4x \quad \text{--- (2)}$$

Sub (1) into (2)

$$\left(\frac{x^2}{4}\right)^2 = 4x$$

$$\frac{x^4}{16} = 4x$$

$$x^4 = 64x$$

$$x^4 - 64x = 0$$

$$x(x^3 - 64) = 0$$

$$x = 0, \sqrt[3]{64}$$

$$= 0, 4$$

When $x = 0, y = 0$

$$x = 4, y = 4$$

(ii) $y^2 = 4x$
 $y = \sqrt{4x}$
 $= 2\sqrt{x}$

3

$$y = \frac{x^2}{4}$$

$$\text{Area} = \int_0^4 (2\sqrt{x} - \frac{x^2}{4}) dx$$

$$= \int_0^4 (2x^{\frac{1}{2}} - \frac{x^2}{4}) dx$$

$$= \left[\frac{4x^{\frac{3}{2}}}{3} - \frac{x^3}{12} \right]_0^4$$

$$= \frac{32}{3} - \frac{64}{12} - 0$$

$$= 5\frac{1}{3} \text{ units}^2$$

Section F

3. When $x = r$ $y = \frac{r^2}{a}$

5

$$x = 2r \quad y = \frac{4r^2}{a}$$

$$\text{Volume} = \pi \int_{\frac{r^2}{a}}^{\frac{4r^2}{a}} (ay) dy$$

$$= \pi \left[\frac{ay^2}{2} \right]_{\frac{r^2}{a}}^{\frac{4r^2}{a}}$$

$$= \frac{a\pi}{2} \left[\frac{16r^4}{a^2} - \frac{r^4}{a^2} \right]$$

$$= \frac{15\pi r^4}{2a}$$

Since $V = \frac{5}{6}\pi a^3$

and $h = \frac{4r^2}{a} - \frac{r^2}{a}$

$$= \frac{3r^2}{a}$$

$$\frac{5}{6}\pi a^3 = \frac{15\pi r^4}{2a}$$

$$10\pi a^4 = 90\pi r^4$$

$$r^2 = \frac{a^2}{3}$$

$$h = \frac{3}{a} \left(\frac{a^2}{3} \right)$$
$$= a$$